

The Betrayal At Cloud City: An Empirical Analysis Of Cloud-Based Mobile Backends

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The Betrayal At Cloud City: An Empirical Analysis Of Cloud-Based Mobile Backends

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Cloud backends provide essential features to the mobile app ecosystem, such as content delivery, ad networks, analytics, and more. Unfortunately, app developers often disregard or have no control over prudent security practices when choosing or managing these services. Our preliminary study of the top 5,000 Google Play Store free apps identified 983 instances of N-day and 655 instances of 0-day vulnerabilities spanning across the software layers (OS, software services, communication, and web apps) of cloud backends. The mobile apps using these cloud backends represent between 1M and 500M installs each and can potentially affect hundreds of thousands of users. Further, due to the widespread use of third-party SDKs, app developers are often unaware of the backends affecting their apps and where to report vulnerabilities. This paper presents SkyWalker, a pipeline to automatically vet the backends that mobile apps contact and provide actionable remediation. For an input APK, SkyWalker extracts an enumeration of backend URLs, uses remote vetting techniques to identify software vulnerabilities and responsible parties, and reports mitigation strategies to the app developer. Our findings suggest that developers and cloud providers do not have a clear understanding of responsibilities and liabilities in regards to mobile app backends that leave many vulnerabilities exposed.

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